

Project Title

Product Success Analysis Using Marketing and Performance Data

Dataset Details

For this project, I used a dataset called **Marketing and Product Performance Dataset** from Kaggle.

- **Source:** Kaggle
- **Link:** <https://share.google/me9DqrznIJz9PUWSA>
- **Size:** 500 rows and 15 columns

The dataset includes information about different products like price, discount, marketing spend, ratings, sales, and other performance-related factors.

Target Variable

The main column I focused on is:

- **Success_Percentage**

This tells how successful a product is, and I used it as the output variable for prediction.

Type of Task

This is mainly a **regression problem**, since I am predicting a continuous value (success percentage).

Problem Statement

Companies spend a lot on marketing, pricing, and improving their products, but it's not always clear how much each factor actually affects product success.

In this project, I tried to understand how things like marketing spend, price, discount, ratings, and supply chain efficiency affect product performance. Along with that, I also built a machine learning model to predict the success percentage based on these factors.

Research Questions

1. How does marketing spend affect product success percentage?
2. What is the relationship between price, discount, and monthly sales?
3. Which product categories perform the best in terms of success and sales?
4. How do customer ratings and number of reviews influence product success?
5. Does supply chain efficiency have an impact on sales performance?
6. How do market trends and seasonality affect product success?
7. Which machine learning model gives the best prediction for product success percentage?

Methodology

First, I loaded the dataset and checked all the columns to understand the data. I removed unnecessary columns like the index column and made sure the data was clean.

Then I did some basic preprocessing. For example, I created categories (low, medium, high) for features like marketing spend, price, discount, ratings, and supply chain efficiency so that it becomes easier to analyze.

After that, I performed basic analysis by grouping the data and calculating averages. Based on this, I created tables and graphs for each research question.

For the last question, I applied machine learning models. I split the data into training and testing sets, trained different models, and compared their performance.

Finally, I saved all the outputs, including tables as CSV files and graphs as PDF files.

Models Used

- Linear Regression
- Decision Tree
- Random Forest

These models are simple and suitable for this type of prediction problem.

Evaluation Metrics

To evaluate the models, I used:

- RMSE (Root Mean Squared Error)
- R^2 Score

These help to understand how accurate the predictions are.

Figures and Tables

For each research question, I created one table and one figure:

- **RQ1:** Scatter plot (Marketing Spend vs Success)
- **RQ2:** Heatmap (Price & Discount vs Sales)
- **RQ3:** Bar chart (Category vs Success)
- **RQ4:** Line plot (Ratings vs Success)
- **RQ5:** Box plot (Supply Chain Efficiency vs Sales)
- **RQ6:** Line plot (Market Trends vs Success)
- **RQ7:** Bar chart (Model comparison)

All graphs are saved as **PDF files**, and tables are saved as **CSV files**.